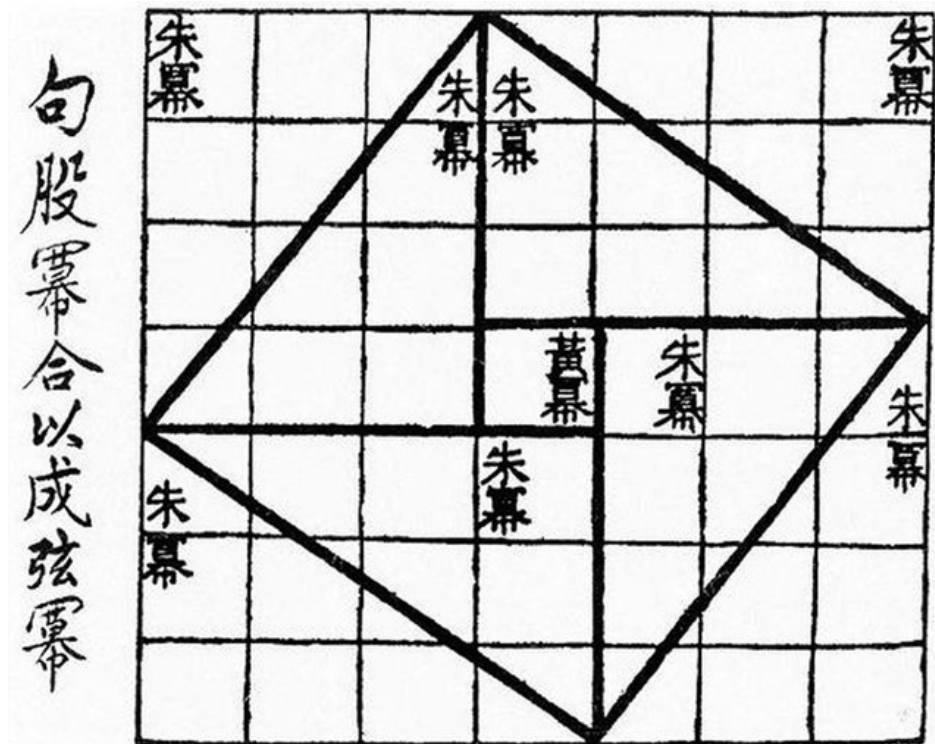




A visual proof of the Pythagorean theorem from the Zhoubi Suanjing.

Dana Ernst, Northern Arizona University

With modifications by Eric Chesebro, Elizabeth Gillaspay, and Cory Palmer (U. Montana)

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Chapter 1

Introduction

1.1 What is mathematics?

What does “math” mean to you? What makes math different than physics or chemistry? than computer science? than a language class?

I would argue that the lowest common denominator of math is logic. In math, there is always an answer to “But why is that true?” Especially in pure math, one’s job as a mathematician is to (1) figure out why something is true, and (2) convince other mathematicians that it’s true.

One can argue that this is the job of any scientist. The difference with (pure) math is that the “But why?” investigation has to happen in our own brains. It doesn’t happen in an external space, like a lab or a computer, where other people can look in and see what’s going on. That means that, regarding point (1), we have to be extra careful, inside our own brains, to be detailed and thorough when figuring out why something is true. Regarding point (2), the arguments we use to convince other mathematicians of the truth we’ve seen in our own brains have to be very precise. Those arguments have to communicate, to someone else’s brain, exactly the same reasoning that we used in our own brain to figure out the truth of a statement.

The focus of this class will be on developing your skills at (1) and (2) above – logical thinking, and mathematical communication. This is a rather different approach to math than you may have taken before. You may love it; you may hate it; you will probably find it frustrating at times. I hope you will also find it exhilarating and satisfying.

1.2 Rules of the Game

Since the goal of this class is for you to improve your skills at logical thinking and mathematical communication, the content of the theorems (and other exercises) that I assign is secondary. My goal in assigning them is to guide you in learning how to construct a proof of a mathematical theorem. In other words: In this class, you will learn to create mathematics. These notes (and I) are here to guide you in developing your own creative abilities, but you are the only one who can do the work.

Because of this, please use your own brain as much as possible. When you have questions, or get stuck and need a hint, please come to me or to your classmates, who are also in the process of becoming mathematical creators. You should *not* look to resources outside the context of this course for help. That is, you should not be consulting the Internet, AI resources, other texts, other faculty, or students outside of our course. On the other hand, you may use each other, the course notes, me, and your own intuition. We will work together to learn how to create mathematics.

Because of this focus on the process, sometimes you won't get to "the end" (a complete proof or solution) by class time. That's OK! It's more important to me that you can repeat the process the next day, than that you can repeat any given proof. So, whether or not you've got a complete solution, please come to class each day prepared to discuss your progress. What did you try, and why? Where did you get stuck? We'll figure out together how to get un-stuck.

1.3 This Class

The majority of the development of your "logical thinking" skills in M 307 will happen outside the classroom. In class, you'll practice your mathematical communication skills. In doing so, you'll sometimes discover that there was a gap in your logical thinking, and so we'll also work on fixing those gaps in class (and on having fewer gaps in future). You'll also learn, in this class, the standard conventions of mathematical writing. For example, mathematicians almost always write in the third person plural: "We now show that the integral of the sum is the sum of the integrals."

The structure of this class is inspired by an educational philosophy called *Inquiry-based learning* or IBL. Loosely speaking, IBL is a student-centered method of teaching mathematics, where the instructor guides the students to construct their own understanding of the subject matter via carefully structured activities, rather than lecture. According to [Laursen and Rasmussen \(2019\)](#), the Four Pillars of IBL are:

- Students engage deeply with coherent and meaningful mathematical tasks.
- Students collaboratively process mathematical ideas.
- Instructors inquire into student thinking.
- Instructors foster equity in their design and facilitation choices.

Because of the IBL philosophy underlying this class, the daily homework in this class will usually not be a simple or straightforward application of concepts that have previously been explained to you. Rather, you will have to think hard and creatively to see how the tools you've developed can be used to solve the homework problems. Often, you will learn new concepts by doing the homework.

1.4 Some advice

You could view these notes as a mountaineering guidebook. I have provided a list of mountains to summit, sometimes indicating which trailhead to start at or which trail to follow. There will always be multiple routes to the top, some more challenging than others. Some summits you will attain quickly and easily, while others might require a multi-day expedition. Oftentimes, your journey will be laced with false summits. Some summits will be obscured by clouds. Sometimes you will have to wait out a storm, perhaps turning around and attempting another route, or even attempting to summit on a different day after the weather has cleared. The strength, fitness, and endurance you gain along the way will allow you to take on more and more challenging, and often beautiful, terrain. Do not forget to take in the view from the top! The joy you feel from overcoming obstacles and reaching each summit under your own will and power has the potential to be life changing. But make no mistake, the journey is vastly more important than the destination.

Because you will be “learning by doing” throughout this class, you will make mistakes. You will, not infrequently, be wrong, or confused, or frustrated. Making mistakes is a central part of learning – if you do everything right the first time, you haven’t really learned anything new that you didn’t know before at some level. Rather than “do it right,” make it your goal to “make new mistakes,” so that you’re always learning something new.

Try to be patient with yourself, and your classmates, as you learn and make mistakes. Try not to give up when you’re feeling stuck and frustrated. Learn to persevere even when you’re not sure what the next step should be. Try something! If it doesn’t work, figure out why, then try something else.

Solving genuine problems is difficult and takes time. You shouldn’t expect to complete each problem in 10 minutes or less. Sometimes, you might have to stare at the problem for an hour before even understanding how to get started.

This class requires both time and persistence, but if you put in the time and the work, it will pay off. The feeling of pride and exhilaration when you’ve solved a really challenging problem yourself is unbeatable. So, hang tough through the confusion; the harder you have to work to figure it out, the more satisfaction you’ll feel when you do. This process is an important part of both everyday life and mathematics.

I am here to guide you as you learn for yourself. I don’t want to spoil the challenge, and I do want you to develop your own “mental muscles” of logical deduction. This means that, while I will happily give you hints if you’re stuck on an exercise, I will always try to give the weakest hint possible that allows you to move forward.

1.4.1 How to use these notes

Discussion of new topics is typically kept at a minimum and there are very few examples in this book. This is intentional. One of the objectives of the items labeled as **Problem** is for you to produce the examples needed to internalize unfamiliar concepts.

1.4.2 Some guidance on proof-writing

Especially in the opening sections, it will not be clear what facts from your prior experience in mathematics you are “allowed” to use. Unfortunately, addressing this issue is difficult and is something we will sort out along the way. In addition, you are likely unfamiliar with how to structure a valid mathematical proof. So that you do not feel completely abandoned, here are some guidelines to keep in mind as you get started with writing proofs.

- The statement you are proving should be written on the same page as the beginning of your proof. If the statement in the book is called a Theorem/Corollary/Lemma/etc, it should have the same label in your writing.
- You should indicate where the proof begins by writing “*Proof.*” at the beginning.
- Make it clear to yourself and the reader what your assumptions are at the very beginning of your proof. Typically, these statements will start off “Assume...”, “Suppose...”, or “Let...”. Sometimes there will be some implicit assumptions that we can omit, but at least in the beginning, you should get in the habit of clearly stating your assumptions up front.
- There will be times when we will need to do some basic algebraic manipulations. You should feel free to do this whenever the need arises. But you should show sufficient work along the way. In addition, you should organize your calculations so that each step follows from the previous. The order in which we write things matters. You do not need to write down justifications for basic algebraic manipulations (e.g., adding 1 to both sides of an equation, adding and subtracting the same amount on the same side of an equation, adding like terms, factoring, basic simplification, etc.).
- On the other hand, you do need to make explicit justification of the logical steps in a proof. As stated above, you should cite a previous definition, theorem, etc. when necessary.
- Carefully consider the order in which you write your proof. Each sentence should follow from an earlier sentence in your proof or possibly a result you have already proved.
- Proofs should be written in complete sentences. You should break sections of a proof into paragraphs and use proper grammar. There are some pedantic conventions for doing this that will be pointed out along the way. This may be different than the way you are used to approaching math homework, but you will get the hang of it.
- You will often want to refer to an earlier definition, problem, theorem, or corollary. In this case, you should reference the statement by number, but it is also helpful to the reader to summarize the statement you are citing. For example, you might write something like, “According to Theorem ??, the sum of two consecutive integers is odd, and so...” or “By the definition of divides (Definition ??), it follows

that...”. One thing worth pointing out is that if we are citing a definition, theorem, or problem by number, we should capitalize “Definition”, “Theorem”, or “Problem”, respectively (e.g., “According to Theorem ??...”). Otherwise, we do not capitalize these words (e.g., “By the definition of divides...”).

- Similar to making it clear where your proof begins, you should indicate where it ends. It is common to conclude a proof with the standard “proof box” ($\square$ or $\blacksquare$).

If you would like additional guidance before you dig in, look over the guidelines in Appendix A: Elements of Style for Proofs. It is suggested that you review this appendix occasionally as you progress through the book as some guidelines may not initially make sense or seem relevant. Be prepared to put in a lot of time and do all the work. Your effort will pay off in intellectual development. Now, go have fun and start exploring mathematics!

1.5 Questions and Observations

It is of utmost importance that you work to understand every proof. (Every!) Questions are often your best tool for determining whether you understand a proof. Therefore, here are some sample questions that apply to any proof that you should be prepared to ask of yourself or the presenter:

- What method(s) of proof are you using?
- What form will the conclusion take?
- How did you know to set up that [equation, set, whatever]?
- How did you figure out what the problem was asking?
- Was this the first thing you tried?
- Can you explain how you went from this line to the next one?
- What were you thinking when you introduced this?
- Could we have ...instead?
- Would it be possible to ...?
- What if ...?

Another way to help you process and understand proofs is to try and make observations and connections between different ideas, proof statements and methods, and to compare approaches used by different people. Observations might sound like some of the following:

- When I tried this proof, I thought I needed to ... But I didn't need that, because ...

- I think that ... is important to this proof, because ...
- When I read the statement of this theorem, it seemed similar to this earlier theorem. Now I see that it [is/isn't] because ...

Chapter 2

A Taste of Number Theory

In this section, we will work with the set of integers,

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

The purpose of this section is to get started with proving some theorems about numbers and study the properties of $\mathbb{Z}$. While you are all very familiar with the integers, in this chapter as well as the rest of the course, as far as possible, *please use only the definitions and theorems provided in the notes*. You might use your prior knowledge for intuition, but intuition is a squirrely creature, very hard to pin down in proofs. Definitions, on the other hand, are carefully and precisely written down, so everyone can agree on exactly what they mean.

It is important to note that we are diving in head first here. There are going to be some subtle issues that you will bump into and our goal will be to see what those issues are, and then we will take a step back and start again. See what you can do!

We use the symbol “ $\in$ ” as an abbreviation for the phrase “is an element of” or sometimes simply “in.” For example, the mathematical expression “ $n \in \mathbb{Z}$ ” means “ n is an element of the integers.”

Definition 2.1. An integer n is *even* if $n = 2k$ for some $k \in \mathbb{Z}$.

Definition 2.2. An integer n is *odd* if $n = 2k + 1$ for some $k \in \mathbb{Z}$.

Notice that we framed the definition of “even” in terms of multiplication as opposed to division. When tackling theorems and problems involving even or odd, be sure to make use of our formal definitions and not some of the well-known divisibility properties. For now, you should avoid arguments that involve statements like, “even numbers have no remainder when divided by 2 while odd numbers do have a remainder.”

Our first issue comes up already. Most would agree with the following fact.

Fact 2.3.

1. Every integer is either even or odd but never both.
2. Sums and products of integers are integers.

The problem is that it is not easy to prove this fact from the definition of $\mathbb{Z}$ we've taken. In fact, our definition is not a very good one. It requires that you already know what integers are.

More formal definitions for $\mathbb{Z}$ have the disadvantage that they are surprisingly abstract, complicated, and counterintuitive. For this reason, we will stick to our definition. The problem now is that we are unable to prove some of the most basic facts about the integers and instead must take them for granted. Fact 2.3 can be proved easily from the various formal definitions of $\mathbb{Z}$, but not from ours.

You may assume henceforth that Fact 2.3 is true. You do not need to prove it and you may use it freely, referring to it by its number.

Theorem 2.4. *If $n \in \mathbb{Z}$ then the sum of n and $n + 1$ is odd.*

Theorem 2.5. *If n is an even integer, then n^2 is an even integer.*

For a mathematician, because we need to be as precise as possible with our reasoning and our language, a statement is only true if it is true *all the time*. For example, the statement “I always want chocolate” is not a true statement if there is a single moment in time when I don't want chocolate. In other words, to establish that a statement is false, all you have to do is find one counterexample – one single setting where your statement is not true.

Problem 2.6. *Either prove, or provide a counterexample to, the statement “The sum of an even integer and an odd integer is odd.”*

Question 2.7. *Did Theorem 2.4 need to come before Problem 2.6? Could we have used Problem 2.6 to prove Theorem 2.4? If so, outline how this alternate proof would go. Perhaps your original proof utilized the approach I'm hinting at. If this is true, can you think of a proof that does not rely directly on Problem 2.6? Is one approach better than the other?*

Problem 2.8. *Either prove, or provide a counterexample to, the statement “The product of an odd integer and an even integer is odd.”*

Problem 2.9. *Either prove, or provide a counterexample to, the statement “The product of an odd integer and an odd integer is odd.”*

$$mn = (2s + 1)(2t + 1) = 4st + 2s + 2t + 1 = 2(2st + s + t) + 1.$$

Fact 2.3 implies that $2st + s + t \in \mathbb{Z}$, so mn is odd.

Problem 2.10. *Either prove, or provide a counterexample to, the statement “If at least one of a pair of integers is even then their product is even.”*

Definition 2.11. *An integer n **divides** the integer m , written $n|m$, if and only if there exists $k \in \mathbb{Z}$ such that $m = nk$. We may also say that m **is divisible by** n .*

Question 2.12. *For integers n and m , how are following mathematical expressions different?*

(a) $n|m$ (b) n/m

Among other things, Fact 2.3 implies that addition, subtraction, and multiplication of integers yields integers. We know that this is not true for division of integers. In other words, it is not always true that $n/m \in \mathbb{Z}$ when $m, n \in \mathbb{Z}$. The upshot: when working with $\mathbb{Z}$ it is a good idea to avoid using n/m . When you feel the urge to divide, switch to an equivalent formulation using multiplication. This will make your life easier when proving statements involving divisibility.

One common proof strategy that you will want to employ in (some of) the following exercises, and throughout the semester, is *proof by contradiction*, also sometimes called *reductio ad absurdum*. The idea is that you start out by assuming that a statement (that you want to prove true) is false. Then, by using mathematical logic and the theorems you've already proved, you obtain a patently false statement. For example, you might end up with the statement $0 = 1$. Since we know $0 \neq 1$, if all of your logical steps were correct, it must be that your original assumption (that the statement was false) is wrong. Consequently, the original statement must be true.

Problem 2.13. Let $n \in \mathbb{Z}$. Either prove, or provide a counterexample to, the statement “If 6 divides n , then 3 divides n .”

Problem 2.14. Let $n \in \mathbb{Z}$. Either prove, or provide a counterexample to, the statement “If 6 divides n , then 4 divides n .”

Theorem 2.15. Assume $n, m, a \in \mathbb{Z}$. If $a|n$, then $a|mn$.

A theorem that follows very easily from another theorem is called a **corollary**. Try to quickly prove the next result using the previous theorem. Be sure to cite the theorem in your proof.

Corollary 2.16. Assume $n, a \in \mathbb{Z}$. If a divides n , then a divides n^2 .

It is considered good form to make your arguments as simple as possible. This usually means that if you can establish a fact by giving a specific example then this is preferable to giving a general argument.

It is also considered good form to use our theorems to prove results rather than resorting to first principles. There's no point in reinventing the wheel if we don't have to. If, while writing a proof, you feel like you've argued this way before, then it is likely that you could quote an earlier theorem instead. In other words, the theorem you're currently trying to prove might just be a corollary.

Theorem 2.17. Assume $n, a \in \mathbb{Z}$. If a divides n , then a divides $-n$.

Theorem 2.18. Assume $n, m, a \in \mathbb{Z}$. If a divides m and a divides n , then a divides $m + n$.

Theorem 2.19. Assume $n, m, a \in \mathbb{Z}$. If a divides m and a divides n , then a divides $m - n$.

Problem 2.20. Assume $a, b, m \in \mathbb{Z}$. Determine whether the following statement holds sometimes, always, or never.

<i>If ab divides m, then a divides m and b divides m.</i>

Justify with a proof or counterexample.

Theorem 2.21. *If $a, b, c \in \mathbb{Z}$ are such that a divides b and b divides c , then a divides c .*

The previous theorem is referred to as **transitivity of division of integers**.

Theorem 2.22. *If $n \in \mathbb{Z}$ then 3 divides $n + (n + 1) + (n + 2)$.*

Appendix A

Elements of Style for Proofs

Years of elementary school math taught us that the answer to a math problem is just a single number, “the right answer.” As you progress further, that becomes less and less true. In proof-based courses (such as this one, and many others you will take), mathematics is about discovering proofs and writing them clearly and compellingly.

The following rules apply whenever you write a proof. I may refer to them, by number, in my comments on your homework and exams. Keep these rules handy so that you may refer to them as you write your proofs.

1. **The burden of communication lies on you, not on your reader.** It is your job to explain your thoughts; it is not your reader’s job to guess them from a few hints. You are trying to convince a skeptical reader who doesn’t believe you, so you need to argue with airtight logic in crystal clear language; otherwise the reader will continue to doubt. If you didn’t write something on the paper, then (a) you didn’t communicate it, (b) the reader didn’t learn it, and (c) the grader has to assume you didn’t know it in the first place.
2. **Tell the reader what you’re proving.** The reader doesn’t necessarily know or remember what “Theorem 2.13” is. Even a professor grading a stack of papers might lose track from time to time. Therefore, the statement you are proving should be on the same page as the beginning of your proof. For an exam this won’t be a problem, of course, but on your homework, recopy the claim you are proving. This has the additional advantage that when you study for exams by reviewing your homework, you won’t have to flip back in the notes/textbook to know what you were proving.
3. **Use English words.** Although there will usually be equations or mathematical statements in your proofs, use English sentences to connect them and display their logical relationships. If you look in your notes/textbook, you’ll see that each proof consists mostly of English words.
4. **Use complete sentences.** If you wrote a history essay in sentence fragments, the reader would not understand what you meant; likewise in mathematics you must use complete sentences, with verbs, to convey your logical train of thought.

Some complete sentences can be written purely in mathematical symbols, such as equations (e.g., $a^3 = b^{-1}$), inequalities (e.g., $x < 5$), and other relations (like $5 \mid 10$ or $7 \in \mathbb{Z}$). These statements usually express a relationship between two mathematical *objects*, like numbers or sets. However, it is considered bad style to begin a sentence with symbols. A common phrase to use to avoid starting a sentence with mathematical symbols is “We see that...”

5. **Show the logical connections among your sentences.** Use phrases like “Therefore” or “because” or “if... , then...” or “if and only if” to connect your sentences.
6. **Know the difference between statements and objects.** A mathematical object is a *thing*, a noun, such as a group, an element, a vector space, a number, an ordered pair, etc. Objects either exist or don’t exist. Statements, on the other hand, are mathematical *sentences*: they can be true or false.

When you see or write a cluster of math symbols, be sure you know whether it’s an object (e.g., “ $x^2 + 3$ ”) or a statement (e.g., “ $x^2 + 3 < 7$ ”). One way to tell is that every mathematical statement includes a verb, such as $=$, $\leq$, “divides”, etc.

7. **“=” means equals.** Don’t write $A = B$ unless you mean that A actually equals B . This rule seems obvious, but there is a great temptation to be sloppy. In calculus, for example, some people might write $f(x) = x^2 = 2x$ (which is false), when they really mean that “if $f(x) = x^2$, then $f'(x) = 2x$.”
8. **Don’t interchange $=$ and $\implies$.** The equals sign connects two *objects*, as in “ $x^2 = b$ ”; the symbol “ $\implies$ ” is an abbreviation for “implies” and connects two *statements*, as in “ $a + b = a \implies b = 0$.” You should avoid using $\implies$ in your formal write-ups.
9. **Say exactly what you mean.** Just as the $=$ is sometimes abused, so too people sometimes write $A \in B$ when they mean $A \subseteq B$, or write $a_{ij} \in A$ when they mean that a_{ij} is an entry in matrix A . Mathematics is a very precise language, and there is a way to say exactly what you mean; find it and use it.
10. **Don’t write anything unproven.** Every statement on your paper should be something you *know* to be true. The reader expects your proof to be a series of statements, each proven by the statements that came before it. If you ever need to write something you don’t yet know is true, you *must* preface it with words like “assume,” “suppose,” or “if” (if you are temporarily assuming it), or with words like “we need to show that” or “we claim that” (if it is your goal). Otherwise the reader will think they have missed part of your proof.
11. **Write strings of equalities (or inequalities) in the proper order.** When your reader sees something like

$$A = B \leq C = D,$$

they expect to understand easily why $A = B$, why $B \leq C$, and why $C = D$, and they expect the *point* of the entire line to be the more complicated fact that $A \leq D$. For

example, if you were computing the distance d of the point $(12, 5)$ from the origin, you could write

$$d = \sqrt{12^2 + 5^2} = 13.$$

In this string of equalities, the first equals sign is true by the Pythagorean theorem, the second is just arithmetic, and the *point* is that the first item equals the last item: $d = 13$.

A common error is to write strings of equations in the wrong order. For example, if you were to write “ $\sqrt{12^2 + 5^2} = 13 = d$ ”, your reader would understand the first equals sign, would be baffled as to how we know $d = 13$, and would be utterly perplexed as to why you wanted or needed to go through 13 to prove that $\sqrt{12^2 + 5^2} = d$.

12. **Avoid circularity.** Be sure that no step in your proof makes use of the conclusion!
13. **Don’t assume what you’re trying to prove.** Beginning students often attempt to write “proofs” like the following, which attempts to prove that $\tan^2(x) = \sec^2(x) - 1$:

$$\begin{aligned}\tan^2(x) &= \sec^2(x) - 1 \\ \left(\frac{\sin(x)}{\cos(x)}\right)^2 &= \frac{1}{\cos^2(x)} - 1 \\ \frac{\sin^2(x)}{\cos^2(x)} &= \frac{1 - \cos^2(x)}{\cos^2(x)} \\ \sin^2(x) &= 1 - \cos^2(x) \\ \sin^2(x) + \cos^2(x) &= 1 \\ 1 &= 1\end{aligned}$$

Notice what has happened here: the student *started* with the conclusion, and deduced the true statement “ $1 = 1$.” In other words, they proved “If $\tan^2(x) = \sec^2(x) - 1$, then $1 = 1$,” which is true but highly uninteresting.

Now this isn’t a bad way of *finding* a proof. Working backwards from your goal often is a good strategy *on your scratch paper*, but when it’s time to *write* your proof, you have to start with the hypotheses and work to the conclusion.

14. **Be concise.** Most students err by writing their proofs too short, so that the reader can’t understand their logic. It is nevertheless quite possible to be too wordy, and if you find yourself writing a full-page essay, it’s probably because you don’t really have a proof, but just an intuition. When you find a way to turn that intuition into a formal proof, it will be much shorter.
15. **Introduce every symbol you use.** If you use the letter “ k ,” the reader should know exactly what k is. Good phrases for introducing symbols include “Let $n \in \mathbb{N}$,” “Let k be the least integer such that...,” “For every real number a ...,” and “Suppose that X is a counterexample.”

16. **Use appropriate quantifiers (once).** When you introduce a variable $x \in S$, it must be clear to your reader whether you mean “for all $x \in S$ ” or just “for some $x \in S$.” If you just say something like “ $y = x^2$ where $x \in S$,” the word “where” doesn’t indicate whether you mean “for all” or “some”.

Phrases indicating the quantifier “for all” include “Let $x \in S$ ”; “for all $x \in S$ ”; “for every $x \in S$ ”; “for each $x \in S$ ”; etc. Phrases indicating the quantifier “some” (or “there exists”) include “for some $x \in S$ ”; “there exists an $x \in S$ ”; “for a suitable choice of $x \in S$ ”; etc.

On the other hand, don’t introduce a variable more than once! Once you have said “Let $x \in S$,” the letter x has its meaning defined. You don’t *need* to say “for all $x \in S$ ” again, and you definitely should *not* say “let $x \in S$ ” again.

17. **Use a symbol to mean only one thing.** Once you use the letter x once, its meaning is fixed for the duration of your proof. You cannot use x to mean anything else.
18. **Don’t “prove by example.”** Most problems ask you to prove that something is true “for all”—You *cannot* prove this by giving a single example, or even a hundred. Your answer will need to be a logical argument that holds for *every example there possibly could be*.

19. **Write “Let $x = \dots$,” not “Let $\dots = x$.”** When you have an existing expression, say a^2 , and you want to give it a new, simpler name like b , you should write the new variable on the left, ie, “Let $b = a^2$.” which means, “Let the new symbol b mean a^2 .” This convention makes it clear to the reader that b is the brand-new symbol and a^2 is the old expression they already understand.

If you were to write it backwards, saying “Let $a^2 = b$,” when you’ve already assigned a meaning to a^2 , then your startled reader would ask, “What is b , and what do I do if $a^2 \neq b$?”

20. **Make your counterexamples concrete and specific.** Proofs need to be entirely general, but counterexamples should be absolutely concrete. When you provide an example or counterexample, make it as specific as possible. For a set, for example, you must name its elements, and for a function you must give its rule. Do not say things like “ θ could be one-to-one but not onto”; instead, provide an actual function θ that *is* one-to-one but not onto.
21. **Don’t include examples in proofs.** Including an example very rarely adds anything to your proof. If your logic is sound, then it doesn’t need an example to back it up. If your logic is bad, a dozen examples won’t help it (see rule 18). There are only two valid reasons to include an example in a proof: if it is a *counterexample* disproving something, or if you are performing complicated manipulations in a general setting and the example is just to help the reader understand what you are saying.
22. **Use scratch paper.** Finding your proof will be a long, potentially messy process, full of false starts and dead ends. Do all that on scratch paper until you find a real

proof, and only then break out your clean paper to write your final proof carefully. *Do not hand in your scratch work!*

Only sentences that actually contribute to your proof should be part of the proof. Do not just perform a “brain dump,” throwing everything you know onto the paper before showing the logical steps that prove the conclusion. *That is what scratch paper is for.*

Appendix B

Fancy Mathematical Terms

Here are some important mathematical terms that you will encounter in this course and throughout your mathematical career.

1. **Definition**—a precise and unambiguous description of the meaning of a mathematical term. It characterizes the meaning of a word by giving all the properties and only those properties that must be true.
2. **Theorem**—a mathematical statement that is proved using rigorous mathematical reasoning. In a mathematical paper, the term theorem is often reserved for the most important results.
3. **Lemma**—a minor result whose sole purpose is to help in proving a theorem. It is a stepping stone on the path to proving a theorem. Very occasionally lemmas can take on a life of their own (Zorn’s lemma, Urysohn’s lemma, Burnside’s lemma, Sperner’s lemma).
4. **Corollary**—a result in which the (usually short) proof relies heavily on a given theorem (we often say that “this is a corollary of Theorem A”).
5. **Proposition**—a proved and often interesting result, but generally less important than a theorem.
6. **Conjecture**—a statement that is unproved, but is believed to be true (Collatz conjecture, Goldbach conjecture, twin prime conjecture).
7. **Claim**—an assertion that is then proved. It is often used like an informal lemma.
8. **Axiom/Postulate**—a statement that is assumed to be true without proof. These are the basic building blocks from which all theorems are proved (Euclid’s five postulates, Zermelo-Frankel axioms, Peano axioms).
9. **Identity**—a mathematical expression giving the equality of two (often variable) quantities (trigonometric identities, Euler’s identity).

10. **Paradox**—a statement that can be shown, using a given set of axioms and definitions, to be both true and false. Paradoxes are often used to show the inconsistencies in a flawed theory (Russell’s paradox). The term paradox is often used informally to describe a surprising or counterintuitive result that follows from a given set of rules (Banach-Tarski paradox, Alabama paradox, Gabriel’s horn).
11. **Without loss of generality**—This phrase comes up frequently in proofs where there are multiple cases to consider but the same logic will work for both. For example, if in your proof you need to assume that either m or n is even, you might say “Without loss of generality, we assume m is even.” [If you just changed the labels of m and n , the same logic would work to handle the case when n is even.]

Appendix C

Definitions in Mathematics

It is difficult to overstate the importance of definitions in mathematics. Definitions play a different role in mathematics than they do in everyday life.

Suppose you give your friend a piece of paper containing the definition of the rarely-used word **rodomontade**. According to the Oxford English Dictionary¹ (OED) it is:

A vainglorious brag or boast; an extravagantly boastful, arrogant, or bombastic speech or piece of writing; an arrogant act.

Give your friend some time to study the definition. Then take away the paper. Ten minutes later ask her to define rodomontade. Most likely she will be able to give a reasonably accurate definition. Maybe she'd say something like, "It is a speech or act or piece of writing created by a pompous or egotistical person who wants to show off how great they are." It is unlikely that she will have quoted the OED word-for-word. In everyday English that is fine—you would probably agree that your friend knows the meaning of the rodomontade. This is because most definitions are *descriptive*. They describe the common usage of a word.

Let us take a mathematical example. The OED² gives this definition of *continuous*.

Characterized by continuity; extending in space without interruption of substance; having no interstices or breaks; having its parts in immediate connection; connected, unbroken.

Likewise, we often hear calculus students speak of a continuous function as one whose graph can be drawn "without picking up the pencil." This definition is descriptive. (As we learned in calculus the picking-up-the-pencil description is not a perfect description of continuous functions.) This is not a mathematical definition.

Mathematical definitions are *prescriptive*. The definition must prescribe the exact and correct meaning of a word. Contrast the OED's descriptive definition of continuous with the the definition of continuous found in a real analysis textbook.

A function $f : A \rightarrow \mathbb{R}$ is **continuous at a point** $c \in A$ if, for all $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|x - c| < \delta$ (and $x \in A$) it follows that $|f(x) - f(c)| < \varepsilon$. If f

¹<http://www.oed.com/view/Entry/166837>

²<http://www.oed.com/view/Entry/40280>

is continuous at every point in the domain A , then we say that f is **continuous on A** .³

In mathematics there is very little freedom in definitions. Mathematics is a deductive theory; it is impossible to state and prove theorems without clear definitions of the mathematical terms. The definition of a term must completely, accurately, and unambiguously describe the term. Each word is chosen very carefully and the order of the words is critical. In the definition of continuity changing “there exists” to “for all,” changing the orders of quantifiers, changing $<$ to $\leq$ or $>$, or changing $\mathbb{R}$ to $\mathbb{Z}$ would completely change the meaning of the definition.

What does this mean for you, the student? Our recommendation is that at this stage you memorize the definitions word-for-word. It is the safest way to guarantee that you have it correct. As you gain confidence and familiarity with the subject you may be ready to modify the wording. You may want to change “for all” to “given any” or you may want to change $|x - c| < \delta$ to $-\delta < x - c < \delta$ or to “the distance between x and c is less than δ .”

Of course, memorization is not enough; you must have a conceptual understanding of the term, you must see how the formal definition matches up with your conceptual understanding, and you must know how to work with the definition. It is perhaps with the first of these that descriptive definitions are useful. They are useful for building intuition and for painting the “big picture.” Only after days (weeks, months, years?) of experience does one get an intuitive feel for the ε, δ -definition of continuity; most mathematicians have the “picking-up-the-pencil” definitions in their head. This is fine as long as we know that it is imperfect, and that when we prove theorems about continuous functions in mathematics we use the mathematical definition.

We end this discussion with an amusing real-life example in which a descriptive definition was not sufficient. In 2003 the German version of the game show *Who wants to be a millionaire?* contained the following question: “Every rectangle is: (a) a rhombus, (b) a trapezoid, (c) a square, (d) a parallelogram.”

The confused contestant decided to skip the question and left with €4000. Afterward the show received letters from irate viewers. Why were the contestant and the viewers upset with this problem? Clearly a rectangle is a parallelogram, so (d) is the answer. But what about (b)? Is a rectangle a trapezoid? We would describe a trapezoid as a quadrilateral with a pair of parallel sides. But this leaves open the question: can a trapezoid have *two* pairs of parallel sides or must there only be *one* pair? The viewers said two pairs is allowed, the producers of the television show said it is not. This is a case in which a clear, precise, mathematical definition is required.

³This definition is taken from page 109 of Stephen Abbott’s *Understanding Analysis*, but the definition would be essentially the same in any modern real analysis textbook.